LAB 9

Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

Objectives

1. To comprehend the fundamental operating principles of a Wireless Local Area Network (WLAN) and master the configuration procedures for wireless routers.
2. To configure the Service Set Identifier (SSID) and implement WPA2 security protocols to establish a secure wireless environment.
3. To establish wireless connectivity between client laptops and the router infrastructure, and validate the connection through network packet analysis using diagnostic tools.

Theory

1. WLAN (Wireless Local Area Network)

A Wireless Local Area Network represents a communication infrastructure that enables multiple computing devices to establish network connectivity without physical cable connections. WLANs utilize electromagnetic radio frequency transmissions to facilitate data exchange between client devices and a central wireless access point or router within a defined geographical area such as residential homes, educational facilities, or corporate environments.

The technology operates primarily within the 2.4 GHz and 5 GHz frequency bands as specified by the IEEE 802.11 standards family. The 2.4 GHz spectrum provides extended range capabilities but operates within a more congested frequency environment, while the 5 GHz band delivers higher data throughput rates with reduced interference but over shorter transmission distances. Wireless access points coordinate all communication by managing channel allocation and controlling medium access among connected devices.

2. SSID (Service Set Identifier)

The Service Set Identifier functions as the unique alphanumeric designation that identifies a specific wireless network within the radio frequency environment. This identifier serves as the network's broadcast name, enabling client devices to differentiate between multiple available wireless networks operating within the same geographical vicinity.

Access points continuously transmit beacon frames containing the SSID information at predetermined intervals, allowing nearby wireless-capable devices to discover and enumerate available network connections. When users initiate connection attempts, their devices reference the SSID to ensure association with the intended network. Network administrators may configure SSID broadcasting as either visible or hidden, with the latter requiring manual network name entry rather than automatic detection.

3. WPA2 (Wi-Fi Protected Access 2)

WPA2 represents the current industry-standard security framework for protecting wireless network communications and controlling access to network resources. This protocol was developed to address security vulnerabilities identified in earlier wireless security implementations by mandating the use of Advanced Encryption Standard (AES) cryptographic algorithms for data protection.

The protocol supports two primary authentication modes: WPA2-Enterprise, which integrates with centralized authentication infrastructure through RADIUS servers, and WPA2-Personal, which utilizes a shared passphrase known to all authorized users. In this laboratory implementation, WPA2-Personal mode was selected, requiring clients to provide the correct pre-shared key during the authentication phase to gain network access authorization.

Network Topology

The laboratory network architecture was constructed using a single WRT300N wireless router designated as Wireless Router0, functioning as the central wireless access point and network coordinator. Two Laptop-PT client devices, identified as Laptop0 and Laptop1, served as the wireless client stations. The router transmitted radio frequency signals to establish communication links with both laptop devices, creating an infrastructure-mode wireless network topology where all inter-client communication is routed through the central access point.

Configuration

The Wireless Router was configured to operate as a Dynamic Host Configuration Protocol (DHCP) server, enabling automatic IP address allocation to wireless clients upon successful network association.

IP Configuration Table

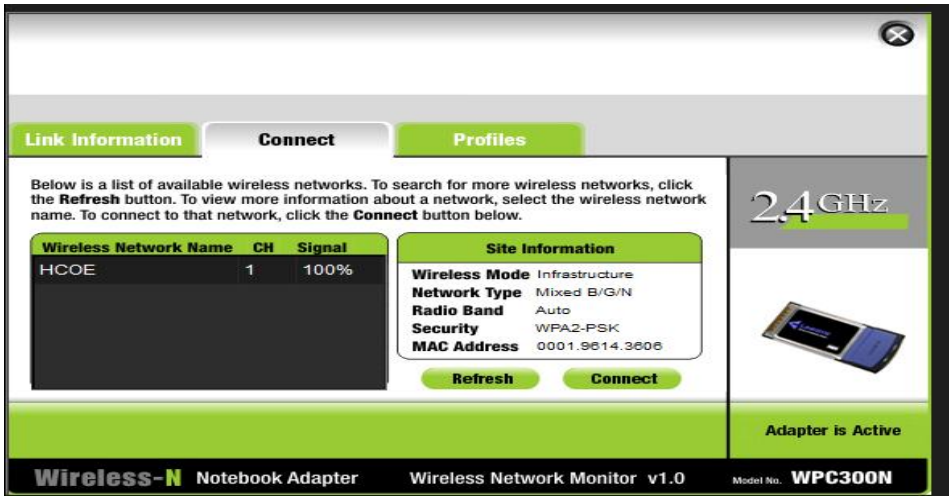
Device	Interface	IP Address	Subnet Mask	Default Gateway	Allocation Method
Wireless Router0	LAN / Wireless	192.168.0.1	255.255.255.0	-	Static
Laptop0	Wireless0	192.168.0.102	255.255.255.0	192.168.0.1	DHCP
Laptop1	Wireless0	192.168.0.103	255.255.255.0	192.168.0.1	DHCP

Procedure

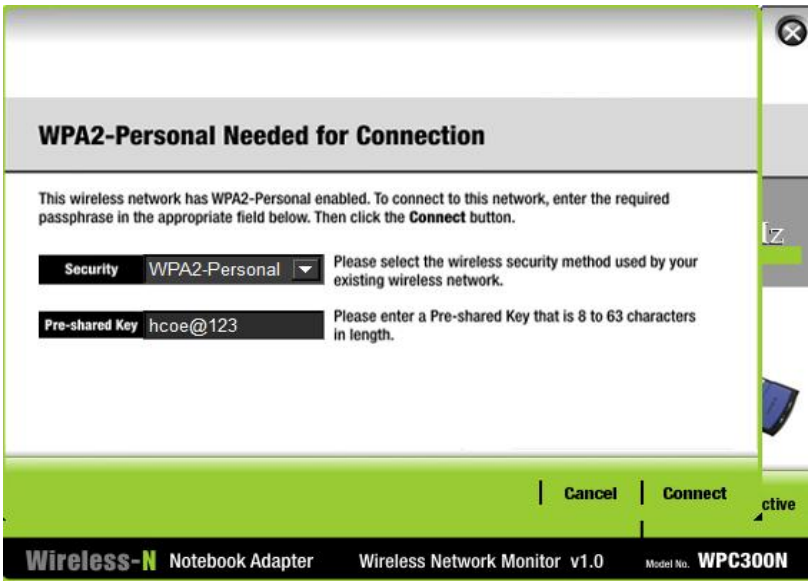
1. Router Setup: The wireless router's graphical configuration interface was accessed through the GUI tab. Within the Wireless Settings section, the network SSID was configured as "HCOE".

Under the Wireless Security configuration panel, WPA2-Personal was selected as the security protocol, and the pre-shared key "hcoe@123" was entered as the authentication passphrase.

2. **Client Connection:** On Laptop0, the PC Wireless software application was launched and the Connect tab was selected. The application performed an active scan for available wireless networks within radio range. The "HCOE" network was successfully detected broadcasting at 100% signal strength on wireless Channel 1.



3. **Authentication:** The "HCOE" network entry was selected from the available networks list and the Connect button was activated. A WPA2-Personal authentication dialog appeared requesting the network's pre-shared key. The passphrase "hcoe@123" was entered to complete the authentication handshake and establish the encrypted wireless connection.

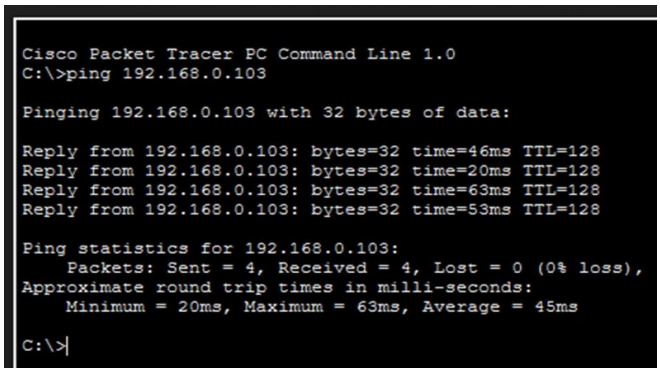


4. IP Assignment: Following successful wireless authentication, both Laptop0 and Laptop1 automatically transmitted DHCP discover messages to request IP configuration parameters. The router's DHCP service responded by allocating unique IP addresses to each client device from the configured address pool.

Testing

Connectivity Verification

To validate end-to-end wireless connectivity and confirm proper packet transmission across the network infrastructure, a connectivity test was performed using the Internet Control Message Protocol (ICMP). Laptop1 had been assigned the IP address 192.168.0.103 through the DHCP allocation process. The Command Prompt terminal was opened on Laptop0 and the command "ping 192.168.0.103" was executed to initiate the connectivity test.



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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:

Reply from 192.168.0.103: bytes=32 time=46ms TTL=128
Reply from 192.168.0.103: bytes=32 time=20ms TTL=128
Reply from 192.168.0.103: bytes=32 time=63ms TTL=128
Reply from 192.168.0.103: bytes=32 time=53ms TTL=128

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 63ms, Average = 45ms

C:\>
```

The ping test successfully generated four ICMP echo request packets and received corresponding echo reply messages from the target device. Response times ranged from 20 milliseconds to 63 milliseconds with an average round-trip latency of 45 milliseconds. The test statistics indicated zero packet loss (0%), confirming that the wireless network infrastructure was functioning correctly and that bi-directional communication was established between the client devices through the wireless router.

Discussion and Conclusion

This laboratory exercise successfully demonstrated the configuration and verification of a functional wireless network environment using Cisco Packet Tracer simulation software. A WRT300N wireless router was configured with the SSID "HCOE" and secured using WPA2-Personal encryption with the pre-shared key "hcoe@123". Both client laptop devices successfully detected the broadcasted network identifier, completed the WPA2 authentication process, and established secure wireless associations. The router's integrated DHCP functionality automatically provisioned each client with appropriate IP configuration parameters. Network connectivity was validated through ICMP ping testing, which demonstrated reliable packet transmission with zero loss rates and acceptable latency characteristics. The laboratory was completed successfully, providing practical experience in wireless network deployment, security configuration, and connectivity verification methodologies.